

Divide and Conquer

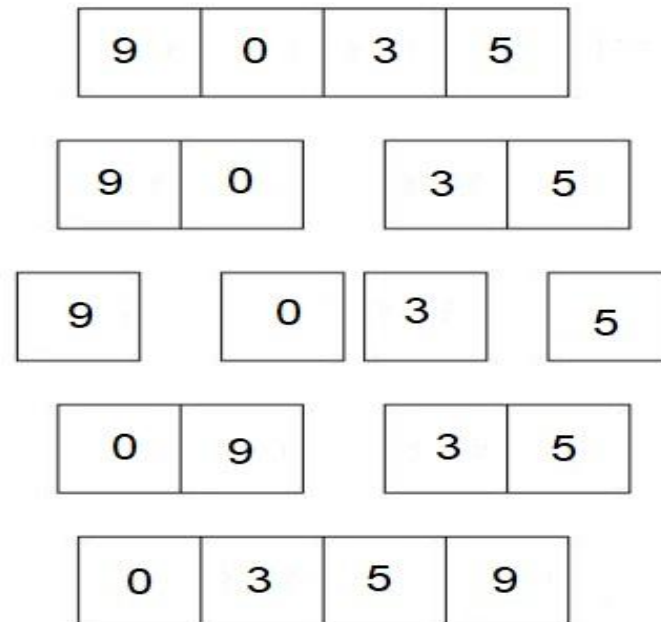
- Given a problem of size n
- Divide into a subproblems
- Each with size $\frac{n}{b}$
- Where $a \geq 1$ and $b > 1$
- Solve each subproblem recursively
- Combine the solutions
- Running time: $T(n) = aT\left(\frac{n}{b}\right) + [merge]$

Merge sort

- **Input:** array $A[1 \dots n]$ of elements
- **Output:** permutation $B[1 \dots n]$ of A such that $B[1] \leq B[2] \leq \dots B[n]$
- Example: $A = [5, 1, 4, 0, 2] \rightarrow B = [0, 1, 2, 4, 5]$
- Applications
 - Organize Files in Folder
 - Contacts in Phone
 - Finding Median
 - Binary Search
 - Data Compression
 - Computer Graphics

Merge sort

- Divide array into two half
- Continue to divide until we are left with one element → Trivially sorted
- Merge elements in sorted order



Merge Operation

2 Subarrays to merge: [2,7,13,20] and [1,9,11,12]

Left subarray



Right subarray



Merged result

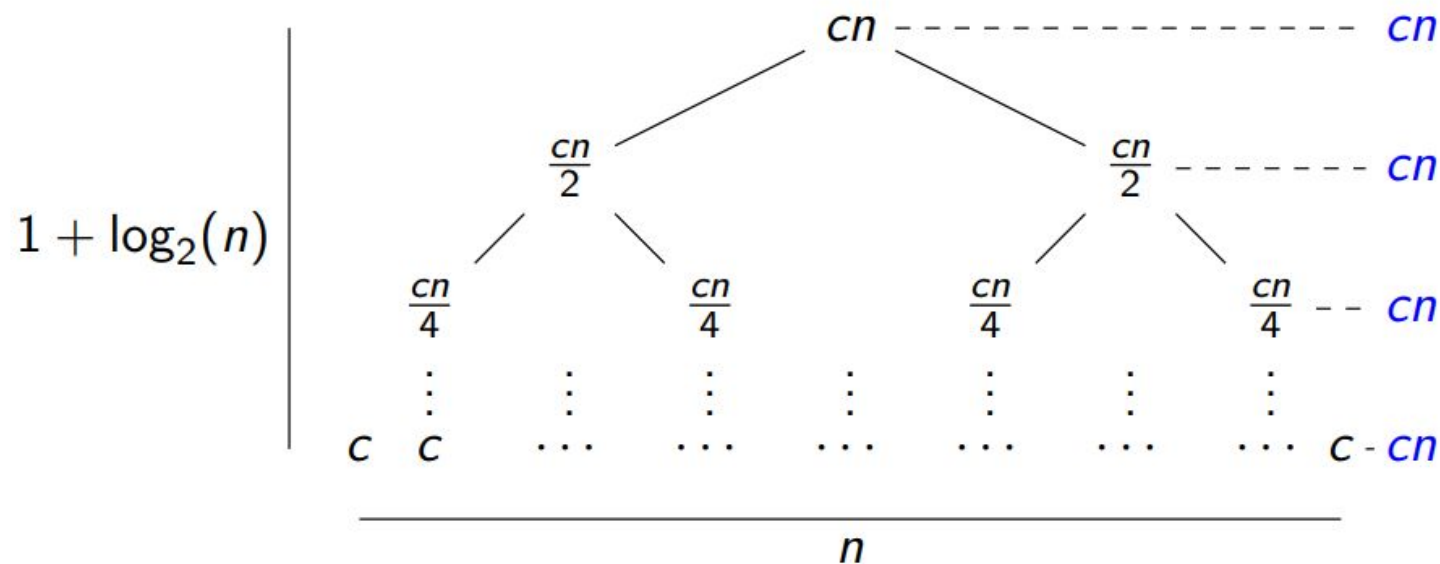


 L = taken from left array

 R = taken from right array

Solving Recurrences

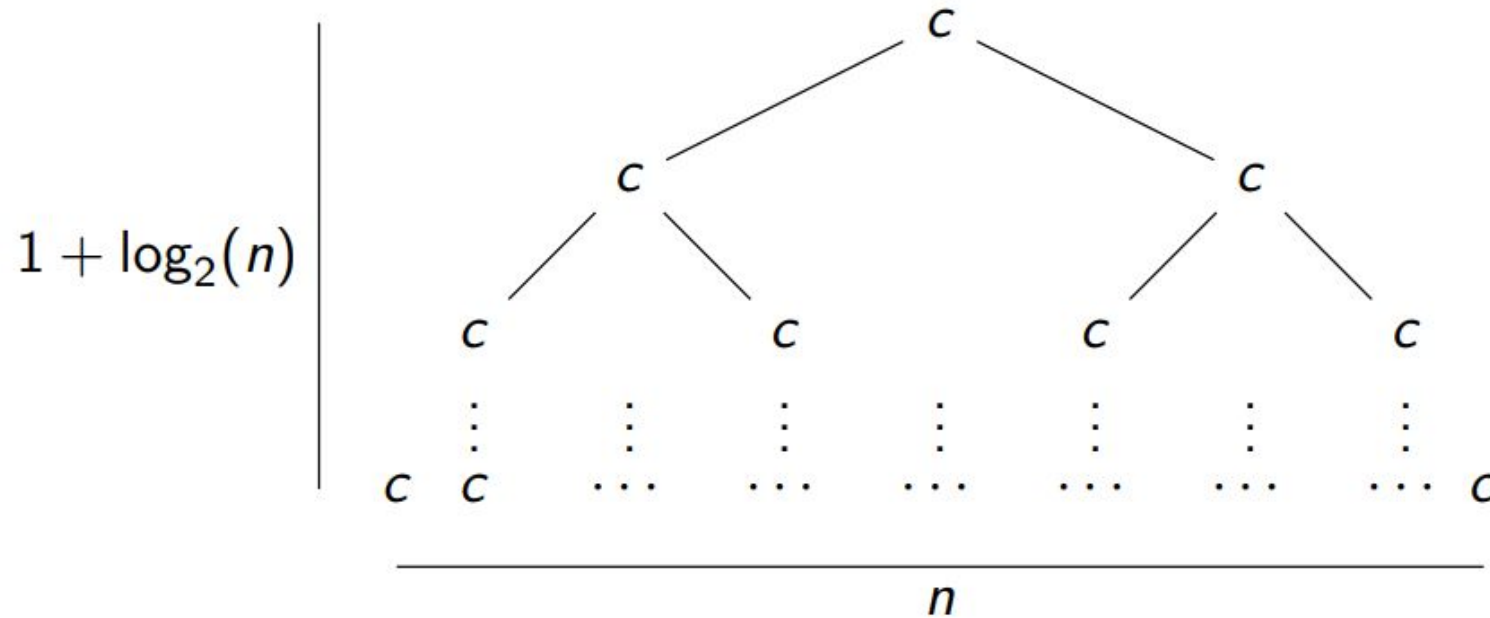
- Solve $T(n) = 2T\left(\frac{n}{2}\right) + cn$, where $c > 0$ is constant
- Recursion Tree



- Equal amount of work done in each level
- Total: $cn \times (1 + \log_2(n)) = cn + cn \log_2(n) = \Theta(n \log_2(n))$

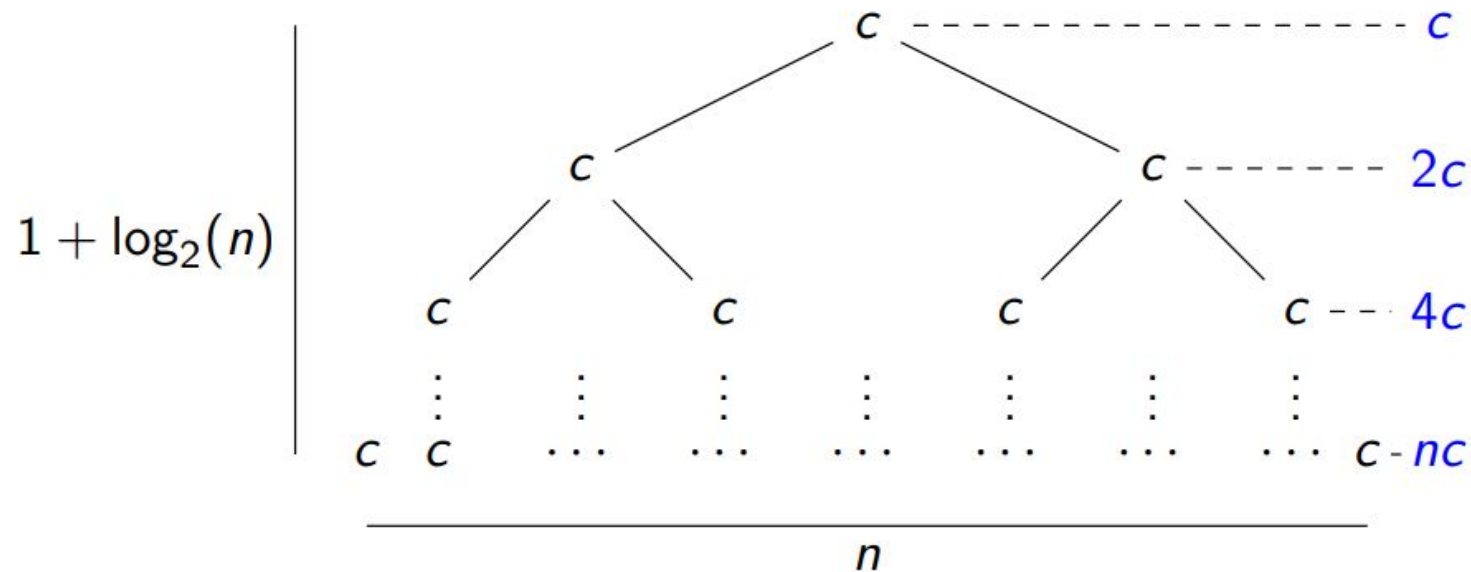
Solving Recurrences

- Solve $T(n) = 2T\left(\frac{n}{2}\right) + c$, where $c > 0$ is constant



Solving Recurrences

- Solve $T(n) = 2T\left(\frac{n}{2}\right) + c$, where $c > 0$ is constant



- Most of the work done in leaves

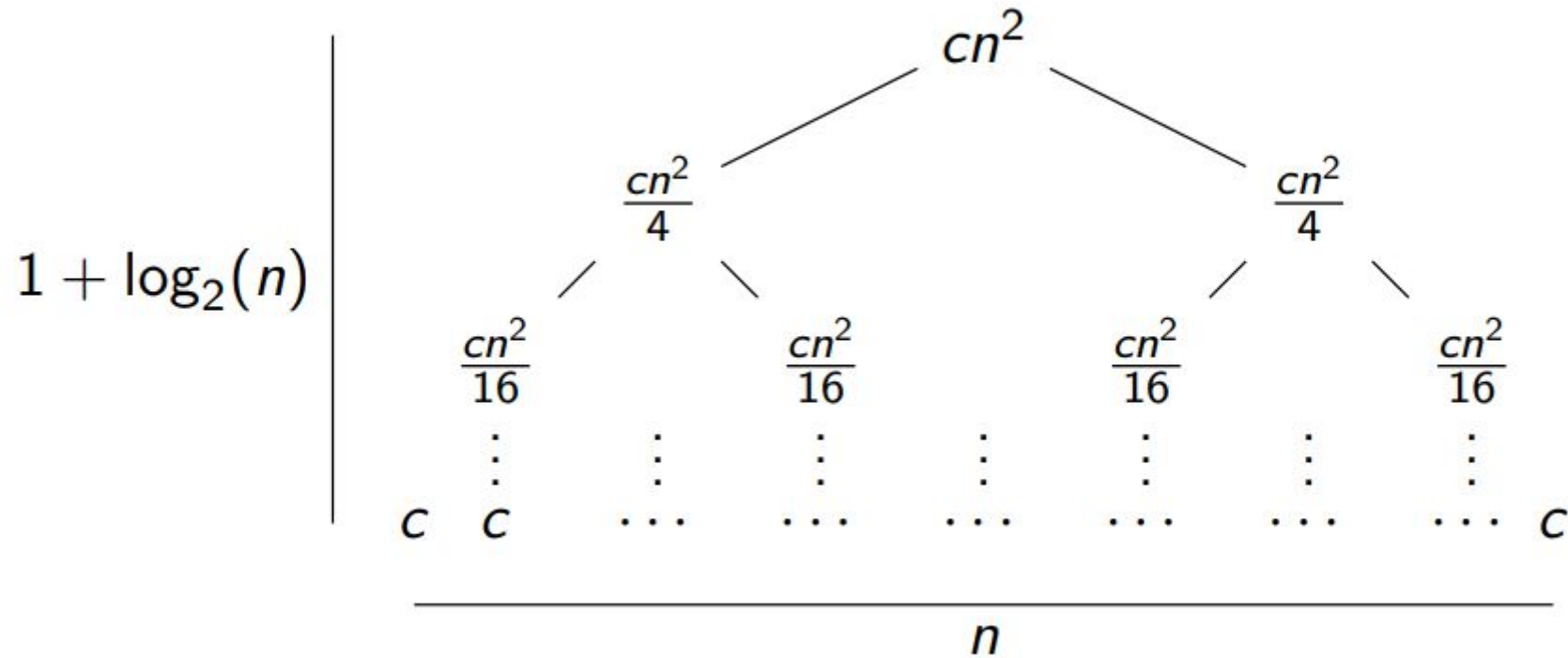
- $T(n) = nc + \frac{nc}{2} + \frac{nc}{4} + \cdots + \frac{nc}{n/2} + \frac{nc}{n}$

$\Rightarrow T(n) = nc \left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{n/2} + \frac{1}{n} \right)$

$\Rightarrow T(n) = \Theta(n)$

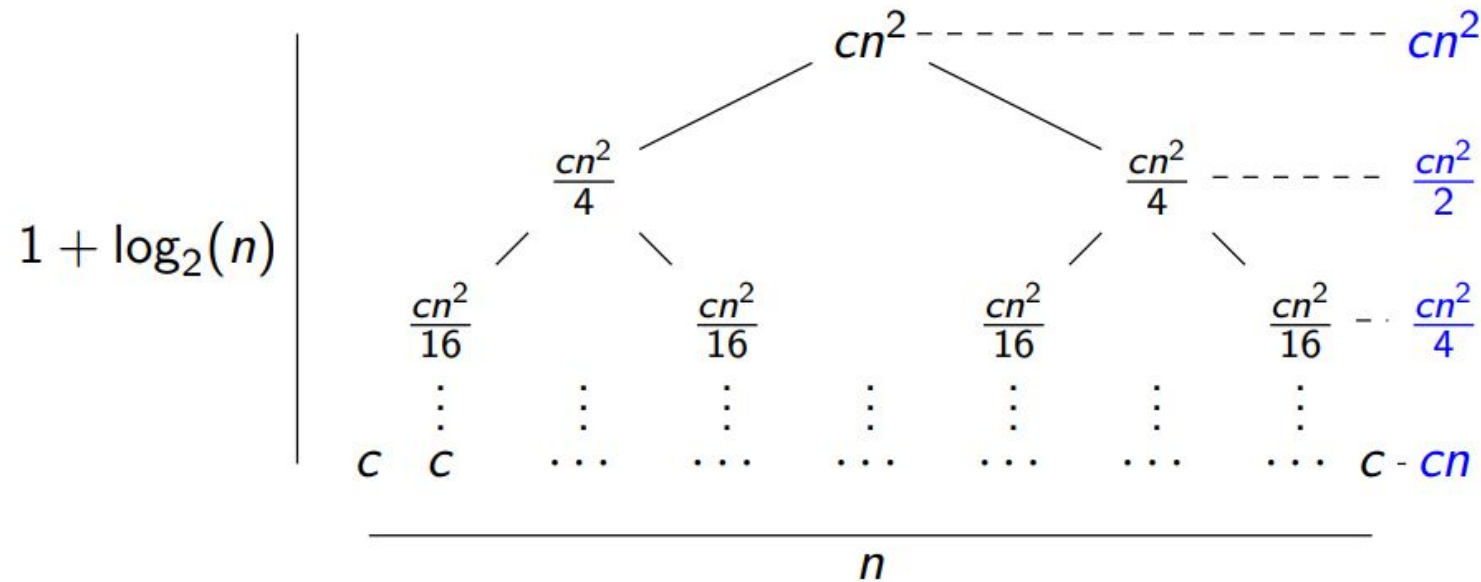
Solving Recurrences

- Solve $T(n) = 2T\left(\frac{n}{2}\right) + cn^2$, where $c > 0$ is constant



Solving Recurrences

- Solve $T(n) = 2T\left(\frac{n}{2}\right) + cn^2$, where $c > 0$ is constant



- Most of the work done in root
- $T(n) = cn^2 + \frac{cn^2}{2} + \frac{cn^2}{4} + \dots + \frac{cn^2}{n}$
- $\Rightarrow T(n) = cn^2 \left(1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{n}\right)$
- $\Rightarrow T(n) = \Theta(n^2)$

Solving Recurrences (Master Theorem)

$$T(n) = aT(n/b) + f(n) \text{ where } a \geq 1 \text{ and } b > 1$$

There are following three cases:

1. If $f(n) = O(n^c)$ where $c < \log_b a$ then $T(n) = \Theta(n^{\log_b a})$
2. If $f(n) = \Theta(n^c)$ where $c = \log_b a$ then $T(n) = \Theta(n^c \log n)$
3. If $f(n) = \Omega(n^c)$ where $c > \log_b a$ then $T(n) = \Theta(f(n))$